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To cite this article: Yunan Zhang & Yongcan Liu (05 Aug 2025): Tame the beast of ChatGPT: developing design principles to strategically integrate ChatGPT into EFL writing through an activity theory perspective, Computer Assisted Language Learning, DOI: [10.1080/09588221.2025.2539156](https://doi.org/10.1080/09588221.2025.2539156)

To link to this article: <https://doi.org/10.1080/09588221.2025.2539156>



Published online: 05 Aug 2025.



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Tame the beast of ChatGPT: developing design principles to strategically integrate ChatGPT into EFL writing through an activity theory perspective

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ABSTRACT

Despite the growing interest in the potential of ChatGPT in writing, empirical research on ChatGPT-assisted English as a Foreign Language (EFL) writing strategies is scarce. This study addresses the central question of how ChatGPT can be strategically integrated into writing, which has been conceptualised as the resolution of contradictions through the lens of Activity Theory (AT). Employing a Design-Based Research (DBR) approach, we have developed an intervention with design principles to guide ChatGPT-assisted writing through three iterative cycles of implementation and revision. In each round, screen recordings of four Chinese EFL writers' ChatGPT-assisted writing, semi-structured interviews on the intervention and AT framework, and stimulated-recall interviews on contradictions and solution strategies were collected and analysed using Reflexive Thematic Analysis (RTA). Four contradictions manifested themselves, wherein the efforts to solve contradictions resulted in the iterative development of the intervention, AT framework, and design principles. Underpinned by the AT framework, four intersected design principles, each consisting of two to three strategies, have emerged from the process of resolving ChatGPT-related contradictions in the AT framework. The four design principles with nine strategies could guide the strategic integration of ChatGPT into EFL writing and have implications for self-regulated learning, AI literacy, and EFL writing instructions.

ARTICLE HISTORY

Received 9 September 2024

Accepted 21 July 2025

KEYWORDS

ChatGPT-assisted L2 writing; AI-Assisted L2 writing; activity theory; design principles; sociocultural theory

Introduction

Since the release of ChatGPT in 2021, recent years have witnessed a growing recognition of the potential of ChatGPT as a writing assistant (Barrot, 2023). Although the field of ChatGPT-assisted writing is in its infancy, some researchers have conducted experiments to examine ChatGPT's effectiveness in writing (Li et al., 2023; Song & Song, 2023; Zhang et al., 2023) or naturalistic studies on learners' writing process and perceptions of ChatGPT (Yan, 2023; Zou & Huang, 2023). These studies have thrown light on the preexisting practices of ChatGPT before-, during-, and

after-writing. However, the tool's effectiveness ultimately depends on learners' ability to skillfully use ChatGPT, while they do not necessarily possess AI literacy (Warschauer et al., 2023; Watson, 2023; ; Wu et al., 2023). Merely explanatory or descriptive knowledge may not suffice in taming the beast of ChatGPT, calling for developing actionable and advisory knowledge to strategically integrate ChatGPT into EFL writing (Cardon et al., 2023; Godwin-Jones, 2022). Moreover, developing ChatGPT-assisted writing strategies involves a trial-and-error process of crafting prompts, necessitating the iterative development of an intervention. So far, no research has explored the strategic integration of ChatGPT into EFL writing through iterative cycles of testing, evaluation, and modification, which could manifest a human-centered AI use amidst concerns about writers' reliance on AI.

Activity theory (AT), as a situative theory of learning, offers a promising framework to conceptualise the strategic integration of ChatGPT into writing as the resolution of contradictions in the sociocultural contexts of writing. Within AT, the research employs a Design-Based Research (DBR) approach to address the research gap through iteratively developing design principles for ChatGPT-assisted EFL writing. In this study, 'design principles' are viewed as guidelines and advice meant to be considered and tried while acknowledging the uniqueness of each situation and the need for local adaptation (Bakker, 2018). Through three iterations, four design principles consisting of nine strategies have been developed, underpinned by an activity system, which may have implications for self-regulated learning, AI literacy, and EFL writing instructions.

AI-assisted writing

AI-assisted writing research has expanded notably in recent years, especially concerning ChatGPT as a writing assistant. Built on large datasets and deep machine learning, ChatGPT could allow for the development of various writing constructs (e.g. language form, style, and structure) (Barrot, 2023; Kasneci et al., 2023), and encourage students to communicate in a low-anxiety environment (Godwin-Jones, 2022; Zhang et al., 2023). Research has shown that ChatGPT could afford writers' idea generation before writing, language and content development during writing, and proofreading and reflection after writing (Su et al., 2023; Warschauer et al., 2023). Nonetheless, ChatGPT may generate plausible but untruthful information and have limitations in assessing some complex writing elements (Godwin-Jones, 2022). An over-reliance on ChatGPT may limit the development of critical thinking and problem-solving skills (Barrot, 2023; Kasneci et al., 2023). The recent discussions on ChatGPT's potential and pitfalls in EFL writing have spotlighted the need to make the best use of ChatGPT (Zou & Huang, 2023).

However, of the current research, strategies for ChatGPT-assisted writing remain under-explored. Much is known about its perceived effectiveness, which can be seen as 'after-the-fact' theories (Bakker, 2018), while much less is known about 'before-the-fact' theories, namely, how to use ChatGPT in EFL writing. Researchers examined students' perceptions of ChatGPT (Yan, 2023; Zou & Huang, 2023), assessed writing improvement through pre-tests and post-tests (Li et al., 2023; Song & Song, 2023), and compared ChatGPT against website-based resources (Zhang et al., 2023) or

teacher feedback (Guo et al., 2023). These studies have confirmed ChatGPT's potential affordances and constraints in writing, enriching our understanding of ChatGPT's functionalities. While the tool itself is important, however, its effectiveness depends on how students use it in local contexts, suggesting the importance of students' strategies amidst various contextual factors (Barrot, 2023). This calls for more attention to the dynamics of the writing process, while there is an empirical gap on the ChatGPT-assisted writing process (Yan, 2023). Learners can critically evaluate contextual factors, skillfully prompt AI to generate content, and adapt AI capabilities to meet particular needs (Warschauer et al., 2023; Watson, 2023; Wu et al., 2023). Moreover, students could employ ChatGPT as a supplementary tool, while previous studies mainly frame it as an alternative to current tools. So far, no research has addressed students' strategies for using ChatGPT alongside other resources in the writing process, which can manifest human agency in AI use (Xiao & Zhi, 2023; Yan, 2023; Zou & Huang, 2023).

Beyond the empirical gap, there are also methodological considerations that warrant attention. There has been some descriptive research on preexisting learner strategies of technology use (Kessler, 2020), which has documented existing practices but might not push the boundaries of what is possible with technologies. Surrounding ChatGPT are ethical issues such as writers' over-reliance and writing dishonesty, suggesting the importance of providing guidance for students (Yan, 2023; Zou & Huang, 2023). Therefore, a totally naturalistic approach that analyses the current state without interference (Bakker, 2018) may be insufficient for revealing the strategic practices. There have been a few researcher-led intervention studies on ChatGPT-assisted writing: Song and Song (2023) and Zou and Huang (2023) implemented ChatGPT as an intervention without a focused consideration of prompt design, while ChatGPT's responses vary greatly with different prompts. Given that crafting appropriate prompts is a trial-and-error process (Dang et al., 2022), there is a need to develop interventions through cycles of testing, evaluation, and modification (Watson, 2023; Wu et al., 2023, 4 November). Additionally, Ya, (2023); Zou and Huang (2023) developed a ChatGPT-embedded practicum to introduce to participants how to apply ChatGPT to L2 writing. However, this intervention from social-media videos raises concerns about reliability due to the variable quality of online content and the widespread misuse of ChatGPT (Kasneci et al., 2023). This highlights the need for developing interventions based on literature and refining them through research, which could be approached through DBR, a methodology to iteratively design interventions.

Moreover, there lack a suitable conceptualisation of ChatGPT-assisted writing process that exhibits greater complexity than traditional ones (Er et al., 2021). Guo et al. (2023) conceptualised ChatGPT as a machine feedback provider to provide corrective feedback, while the uniqueness of Generative AI (GenAI) lies in text generation and interactive engagement that can afford various stages of writing (Su et al., 2023; Warschauer et al., 2023). Researchers also applied scaffolding to indicate ChatGPT as a more advanced peer (Song & Song, 2023; Yan, 2023; Zou & Huang, 2023), whereas ChatGPT may not necessarily be more knowledgeable since it often generates unverified information rather than real knowledge (Kasneci et al., 2023). According to Godwin-Jones (2022), AI-assisted writing should be analysed through

the dynamic interactions among the learner, software, and setting, calling for a theoretical perspective that posits the tool in its interplay of various mediating factors. The next section will discuss the chosen framework and the rationale.

Activity theory (AT)

This research adopted cultural-historical AT to conceptualise writing as a collective and situated activity. Initially built on Vygotsky's idea of social mediation, an AT framework consists of three elements, as visualised in the upper triangle in [Figure 1](#): subjects (humans) adopt mediating artifacts (signs and tools) to complete an object (goals and motives) (Engeström, 1987). Leont'ev and Engeström expanded this triad to highlight the sociocultural contexts through three elements below the upper triangle in [Figure 1](#): rules (explicit and implicit norms), community (a group of people sharing the same object), and the division of labor (distribution of tasks in the community) (Engeström, 1987). These elements interact with each other, comprising a collective, mediated, and object-driven activity system as a prime unit of analysis (Engeström, 1987). Within AT, Guo et al. (2023) found students formed a writing community where they served as idea contributors and help-seekers and used various, with the chatbot as a writing tutor. The ChatGPT-assisted writing activity system was further shaped by rules of argumentative writing and other mediating tools beyond ChatGPT (e.g. translation tools and typing assistants) (Guo et al., 2023). Guo et al. (2023) have thus articulated the congruence between AT and the conceptualisation of ChatGPT-mediated writing process as a socially situated, object-oriented activity system. Nevertheless, the complexity of the chatbot-assisted writing might not be fully explored, since the chatbot, as an innovative tool, could transform the activity, which could have been explored through the concept of contradictions, another central construct of AT.

A contradiction is defined as 'a unity of opposites, opposite forces, or tendencies' (p.370) within an activity system (Engeström & Sannino, 2011). According to Engeström (1987), when an activity system incorporates a new component such as a new technology, it will clash with pre-existing elements, possibly resulting in four levels of contradictions: primary contradictions within a single node of the framework, secondary contradictions between two nodes, tertiary contradiction between an old activity system with a new one, and quaternary contradiction with adjacent activities. The different contradictions can produce disturbances and obstacles, but

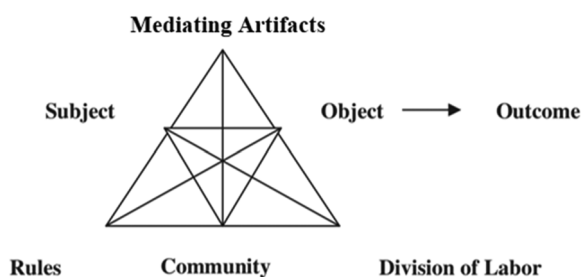


Figure 1. Activity system (Engeström, 1987).

working through contradictions may promote development and innovation (Engeström, 1987). In this context, the introduction of ChatGPT can lead to various contradictions within students' writing activity systems. For instance, ChatGPT could help achieve writing goals, while increased exposure to ChatGPT may lead to over-reliance on technologies and hinder writing development (Warschauer et al., 2023; Watson, 2023; Wu et al., 2023), which suggests a contradiction between the tool and object, whereas the possible contradictions that AI may bring into a ChatGPT-assisted writing activity system have not been addressed in current literature.

More importantly, contradictions can offer possibilities for expansive developmental transformations of the activity system. When contradictions are aggravated, subjects may question the established mode and make collective efforts to solve contradictions, leading to an expansive transformation from the existing mode of activity to a historically new form (Engeström, 1987). For example, Li (2013) applied AT to examine the contradictions EFL writers face. One tension was between writers' novice status and the expectation to write like experts, conceptualised as a within-subject contradiction. Another arose from mismatches between institutional goals and learners' personal objectives, conceptualised as a within-object contradiction. These contradictions demonstrated the challenges writers face and the dynamic adjustments they make to navigate conflicting demands, proving AT as a productive framework for capturing the complexities inherent in EFL writing. Accordingly, the concept of contradictions can not only reveal tensions within an activity system, but illustrate the developmental process of how an innovative form of a ChatGPT-assisted writing activity grows out of the contradictions within the old one.

Another notion central to AT is appropriation, a process through which the subject 'adopts tools available for use...and through this process internalises ways of thinking endemic to specific cultural practices' (Grossman et al., 1999, p. 15). There are different levels of appropriation, demonstrating it as a developmental process from the simple recognition of tools to complete mastery (Grossman et al., 1999). Technology-enhanced writing can be viewed as a mediated appropriation process where writers internalise external mediating resources into psychological tools through strategies such as selecting, evaluating, adapting, and emotion-regulating (Jiang & Yu, 2022; Lei, 2008). Furthermore, given that ChatGPT can be readily adapted by users to fit contextual needs through prompt engineering (Su et al., 2023; Warschauer et al., 2023), students can exert agency beyond merely using mediating resources to participate in the optimisation of writing strategies, wherein students can internalise the way to appropriate ChatGPT and other tools. Therefore, beyond the tool mediation, the concept of appropriation highlights the subjects' internalisation of mediating resources, adding an intrapersonal dimension to AT and appreciating the active role humans play in the activities (Grossman et al., 1999).

Altogether, this research draws upon three main constructs of AT: activity system framework, contradictions, and appropriation. The activity system can depict an overall picture of writing activities, while contradictions may occur within or between the mediating factors after ChatGPT enters the system. As individuals appropriate ChatGPT and other mediating resources to solve contradictions, the present-mode activity system will be driven towards an innovative mode. The evolution of the

activity system then underpins the development mechanism of ChatGPT-assisted writing strategies that finally form some well-argued design principles.

Present study

Within AT, the research conceptualises the strategic integration of ChatGPT into EFL writing as the resolution of contradictions through appropriating mediating resources in the activity system. The research question is: how can EFL writers appropriate mediating resources to solve contradictions emerging from ChatGPT-assisted EFL writing activities? To answer this, we did not compare different participants to justify the most strategic use of ChatGPT, but synthesised appropriation strategies contributed by participants of varied experiences to make the design principles more transferrable. We identified contradictions and detailed writers' appropriation strategies in solving contradictions, addressing the objective: developing design principles to guide the strategic integration of ChatGPT into EFL writing.

Methodology

We adopted a DBR approach, a methodology of designing, developing, and evaluating educational interventions to both solve practical problems and advance our knowledge about the interventions and their development processes (Bakker, 2018). The iterative cycles of design, implementation, and revision in DBR could ensure the strategies of ChatGPT-assisted writing are systematically developed, with researchers and participants cooperatively optimising the intervention. DBR's duality in theory and practice can also develop design principles and an activity system framework to understand writing activities in-depth. Additionally, as opposed to a traditional close-ended DBR that seeks to control variables, an open approach to DBR shares important commonalities with AT, both committing to context-relevant and collaborative research with stakeholders (Bakker, 2018; Engeström & Sannino, 2011).

This study was also a multiple case study that examined an issue by linking together several cases at different sites (Stake, 2017). There are two important dimensions to our case study approach. First, traditional DBR with larger sample sizes in randomised trials was criticised for failing to consider the contextual factors (The Design-Based Research Collective, 2003). By contrast, through a multiple case study design, we focused on individual cases to show 'the value of attending to context' in DBR (The Design-Based Research Collective, 2003, p.7). The iteration of DBR can help increase the transferrability of case studies findings, while the contextuality of case studies can ground design principles in authentic settings (Kelly et al., 2008). Second, although case studies cannot lead to generalisation across all writers, a nuanced understanding of several cases can generate insights relevant to similar contexts, as demonstrated by previous writing research with small sample sizes (e.g. Guo et al., 2023; Li, 2013). This can also lead to 'the continuous refinement and development of a fully developed theory' (Dooley, 2002, p.351), which, in this context, contributed to the development of our AT framework, intervention, and design principles.

Table 1. Participants profiles.

Participants	Age	IELTS Writing Experience	Previous Appropriation Experience of ChatGPT
Wang	21	Wang has taken IELTS exams twice and scored 7 in writing.	Before the research, they seldom used ChatGPT and were unfamiliar with it.
Yan	27	Yan has taken IELTS exams twice and scored 7 in writing	
Zhang	23	Zhang has taken IELTS exams one time and scored 7.5 in writing	Zhang majors in linguistics and studied computational linguistics as a Master's course. He often uses ChatGPT for academic reading and writing.
Li	23	Li scored 25 in TOEFL writing and had experience for IELTS writing	Li is a PhD student in Natural Language Processing (NLP) and often uses ChatGPT for everyday writing.

Research context and sampling

This study situated itself in the Chinese EFL context and chose IELTS writing task 2 as an example task for it is a representative argumentative writing task with established evaluation criteria. Purposive sampling was used to include Chinese EFL student writers of different appropriation levels, forming a representative sample, and to approach individuals who could provide quality information on the topic (Denscombe, 2014). Additionally, an element of convenience was incorporated to select participants based on their availability and willingness to participate (Denscombe, 2014). Four voluntary participants were recruited (Table 1) based on following criteria through our personal contact: 1) having access to ChatGPT 3.5 and willing to use it in EFL writing; 2) proficient in English so that language barriers would not interfere writers’ interactions with ChatGPT 3.5 that functions better in English than other languages (OpenAI, 2024); 3) experienced in EFL argumentative writing to lay foundations for later participation; 4) each two at lower and higher levels of appropriating ChatGPT, which could provide us with richly diverse data. Though at different ages, all four participants were studying postgraduate courses at a university in the UK at the time of data collection. Their writing scores (IELTS 7.0 or above and TOEFL Writing 25) indicate an advanced level of English proficiency, as both scores fall into the ‘good to excellent’ range according to official scoring descriptors. The sample size was small to enable in-depth qualitative analysis, aligning with the exploratory nature of the study.

Research procedure

Figure 2 presents the three main phases of DBR: preparation and design, intervention and enactment, and retrospective analysis (Bakker, 2018). As shown in Figure 2, in phase 1, we developed an initial intervention, two initial design principles and a preliminary activity framework, though these were hypothetical in nature and open to revision. The intervention was a three-stage guideline from Su et al. (2023); Warschauer et al. (2023) who offered detailed suggestions including sample prompts on using ChatGPT in argumentative writing from outline preparation, content drafting, to proofreading. We summarised and adjusted Su et al.’s (2023); Warschauer et al. (2023) suggestions to integrate ChatGPT into IELTS writing. For example, Figure 3 is a screenshot of the initial intervention on instructing writers to use ChatGPT for proofreading with a specific prompt.

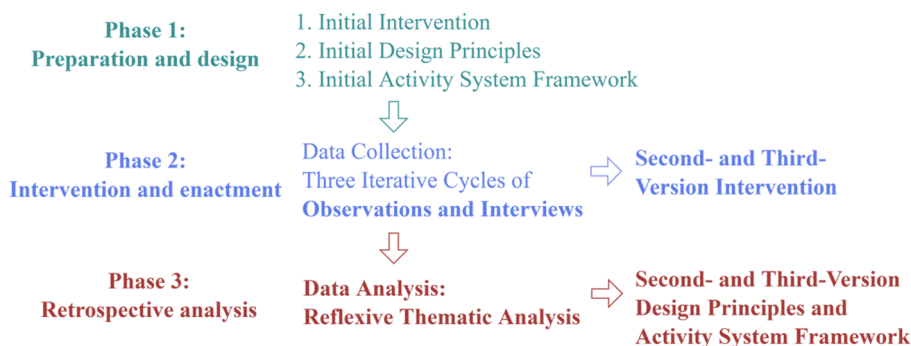


Figure 2. Research procedure.

Stage 3: Proofreading

After writing a whole essay, you can send this prompt to ChatGPT and revise the essay.

“Act as a professional writing tutor who is currently working on giving feedback to my IELTS writing task 2.

Please follow the IELTS writing band descriptors to evaluate and give critical, specific, and constructive feedback to my writing from the following aspects: task response, coherence and cohesion, lexical resource, and grammatical range and accuracy

(Please copy the target score band descriptors here)

I will give you the student writing later.

Please be specific and critical.

Do not start until I provide you with a learner writing.

Do you understand?”

Figure 3. Screenshot of the initial intervention on using ChatGPT for proofreading.

Phase 2 consisted of three iterative cycles of intervention and enactment. Each cycle involved semi-structured observations of the writing process with field notes, a semi-structured interview, and a stimulated-recall interview ([Appendix A. Interview Questions](#)). Each participant engaged with the writing task within their own writing context and without interaction with other participants, thereby representing a relatively distinct and bounded case (Denscombe, 2014). The semi-structured interview was to gather feedback on the intervention and participants’ reflections on the six elements of the activity system. The stimulated-recall interview was to explore contradictions and participants’ appropriation strategies. Interview questions were revised based on a questionnaire by Guo et al (2023) to elicit participants’ reflections on the six elements of the activity system. For example, we adjusted Guo et al. (2023)’s question ‘What do you think are the keys to writing a high-quality argumentative essay?’ to two separate questions on the ‘rule’ and ‘goal’ of the activity to elicit participants’ perceptions. As historically emerging and systematically developing phenomena, contradictions cannot be directly accessed in research, but can be recognised through their manifestations in words and actions (Engeström & Sannino, 2011).

This research approached the inner contradictions from the outer human actions, namely, writers' proactive actions of appropriating ChatGPT for writing as both observed in semi-structured observations and described in interviews. Before writing, participants were asked to notice whether they would encounter any contradictions while writing. During their writing, participants' proactive actions of using ChatGPT instead of following the intervention were noted down because they might suggest the occurrence of contradictions. After writing, chat logs or screenshots of these actions were used as stimuli to investigate contradictions and solution efforts. Each interview, lasting 30 to 45 min, was conducted *via* online meeting software chosen by participants and in their chosen languages (English for Yan and Mandarin for Wang, Zhang, and Li).

In Phase 3, we used Reflexive Thematic Analysis (RTA) to analyse the interview data. The reflexivity of RTA allowed us to utilise prior experiences and theoretical perspectives to enrich the understanding of qualitative data (Braun & Clarke, 2022). Using NVivo, we followed a procedure of familiarisation, transcription, generating initial descriptive codes, grouping them into themes, and refining themes. Regarding the activity system framework, themes from semi-structured interviews were mapped onto the framework to enrich it towards the second and third versions (Figure 2). For example, the transcript that '(I wrote in) the way that I was taught to in my previous schooling experience in argumentative writing' was coded as 'argumentative writing instructions' to update the 'rule' element of AT framework. The design principles in Figure 2 were derived from the stimulated-recall interviews. While coding, we referred to the recordings of participants' mentioned instances to sort out actions comprising the resolution of contradictions. Actions were then synthesised and generalised to form appropriation strategies and design principles. For example, Wang's action of using e-dictionaries was identified and coded as an appropriation strategy of 'consulting E-dictionary for standard vocabulary use' with her relevant explanations in interviews. This strategy was then categorised into the first design principle on orchestrating all available tools. The second- and third-versions of design principles have thus emerged. Screen recordings and chat logs were also triangulated to support the credibility of our analysis.

AI ethics

According to AI ethical guidelines from OpenAI (2024) and the AI Governance Professional Committee in China (2021), this study has considered three fundamental dimensions of AI ethics: beneficence, human criticality, and agency. First, positioning AI tools as an assistant instead of a replacement, the study aims to assist EFL writers. Secondly, human criticality has been centralised throughout the research to minimise inherent bias in LLM. The intervention outlines a critical examination of ChatGPT output, with reminders of ChatGPT's limitations. Thirdly, considering concerns over human dependence on ChatGPT, some safeguards were added to ensure human agency: before writing, participants could merely use ChatGPT to refine their outline. During writing, they could not copy ChatGPT outputs. The screen recordings also helped monitor their honesty in writing, given that participants would be reminded if they copied ChatGPT's output into their writing.

Findings

We conducted three iterative cycles to identify and generalise mediating factors, contradictions, and participants' appropriation strategies, contributing to the development of the intervention, activity system framework, and design principles. This article focuses on how these design principles emerged from the process of resolving contradictions in the activity system and could guide EFL writers' appropriation of ChatGPT. Due to the limits of space and this article's focus, this section does not present the evolution of the intervention in detail, which has been summarised in [Appendix B](#). The first two principles were formed before the research and reinforced during Iteration 1 and 2, while the other two principles were newly identified from Iteration 1. The first iteration involved the major construction of design principles and was the focus of this section. Since participants found the second-version intervention significantly more useful but still encountered some contradictions, some minor refinements were made in the second iteration. In the third iteration, participants widely showed positive attitudes toward the third-version intervention and had few suggestions, which justified the need to conclude the research. The final design principles underpinned by an activity system framework were thus presented as the end product.

Overview of activity system and contradictions

The three rounds of research were conceptualised as three cycles of expansive transformation of the activity system from its present mode to a new one collectively generated by participants through a mediated appropriation process. [Figure 4](#) presents how the activity system transformed from its initial mode to include more elements (marked in red in [Figure 4](#)) in the final framework. In the initial activity system identified before the research, Chinese EFL writers appropriate ChatGPT and the first-version intervention to complete the task. More mediating artifacts including technological tools and signs, more specific rules regarding relevant genres, and ChatGPT's adaptive roles at different writing stages, have been added to the final framework after three iterations, as shown in [Figure 4](#). More importantly, such development is driven by contradictions emerging from the process. Primary contradictions arose within a single component, and secondary contradictions across components, depicted with the dotted-line arrows:

1. Within-tool contradiction: Tensions arose from the difficulty of selecting a suitable tool from the multiple digital tools with similar functions (e.g. ChatGPT, e-dictionary, Automated Writing Evaluation (AWE) tools).
2. Sign-object contradiction: This contradiction emerged as participants felt incapable of completing the object with limited lexical resources (signs).
3. Tool-object contradiction: The writing task demands grammatical accuracy, but the use of ChatGPT brought forth contradictions when ChatGPT gave false grammar revision during proofreading.
4. Subject-object contradiction: Writers, when unfamiliar with the topic, found it challenging to develop adequate content, resulting in tensions with the writing task.

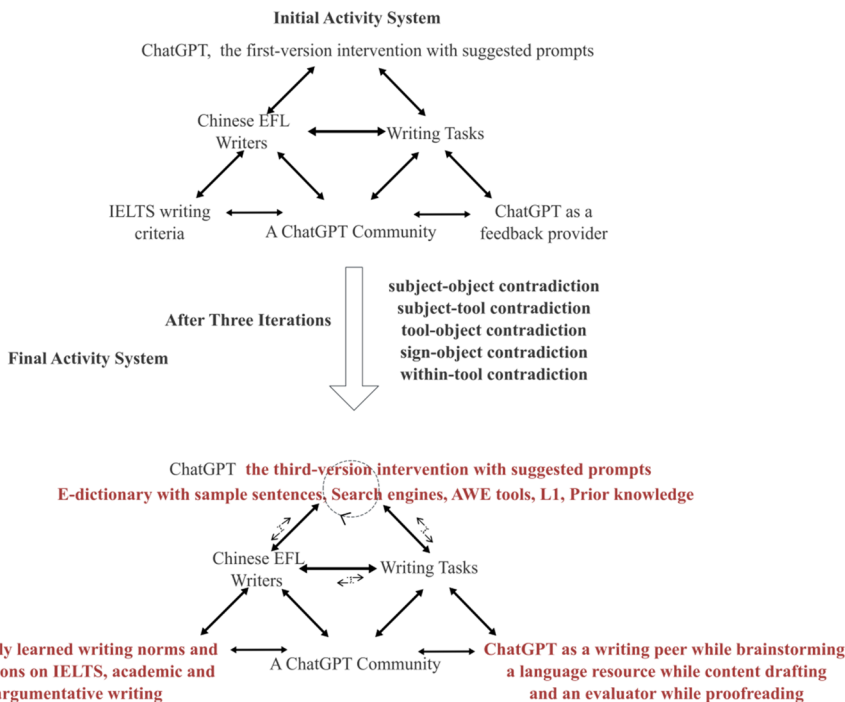


Figure 4. Development of the activity system.

5. Subject-tool contradiction: Writers expected actionable feedback, while ChatGPT sometimes provided overly vague and long responses and thus contradicted writers' needs.

Iteration 1: construction

Design principle 1: orchestrate all available tool

The first principle is to align the functionalities of various tools with writers' specific needs. The principle was formed before implementation and reinforced in this cycle when writers appropriated e-dictionaries in solving a sign-object contradiction.

Appropriation strategy 1-1: consulting E-dictionary for standard vocabulary use. This strategy was derived from the challenge to use limited lexical resources to fulfill the task, as said by Wang: 'I couldn't think out what English expressions to be used...It was the barrier posed by limited vocabulary reservoir'. As illustrated in Figure 5, the current limited vocabulary (signs) could not serve the need to fulfill the writing task (object), causing a sign-object contradiction. This contradiction propelled Wang to introduce e-dictionaries into the system. She thus used e-dictionaries to search for vocabulary at the content drafting stage.

Although ChatGPT has the similar function, Wang chose e-dictionaries over ChatGPT, explaining that: 'I'm used to looking through these sample sentences...I can see how authoritative it (sample sentence) is and then see how the sample sentence is used'. The sample sentences were another mediating resource (signs)

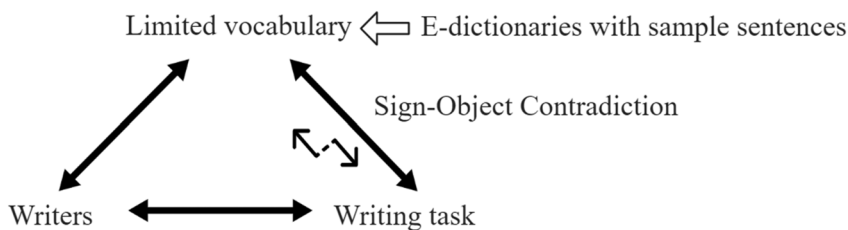


Figure 5. Sign-object contradiction at content drafting stage between writers' limited vocabulary and the writing task.

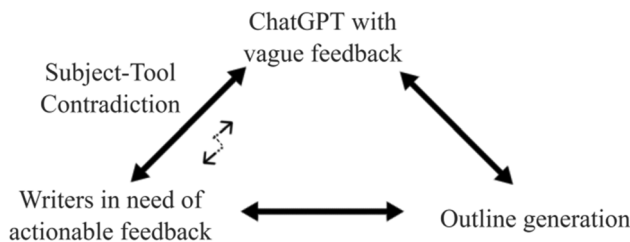


Figure 6. Subject-tool contradiction at outline generation stage (when writers need actionable feedback but meet ChatGPT's vague feedback).

provided by e-dictionaries, informing her of standard vocabulary use (rules). Her use of e-dictionaries showed her appropriation of rules, tools, and signs in solving the sign-object contradiction. The first strategy was generalised to suggest the functionality of e-dictionaries in informing standard language use.

Design principle 2: prompt ChatGPT with specificity

The second principle involves providing clear and detailed instructions to avoid vague responses. Although the initial prompts incorporated specific requirements with established writing criteria, all the participants highlighted the vagueness of ChatGPT's response during the outline generation and proofreading stages. Subject-tool contradictions arose at these stages when ChatGPT's vague feedback could not meet writers' needs. Two following appropriation strategies were included.

Appropriation strategy 2-1: asking for elaboration based on prior knowledge. This strategy was shared among all the participants when they resolved a subject-tool contradiction. As represented in Figure 6, at the outline generation stage, ChatGPT provided overly vague advice and failed to meet writers' need for actionable feedback, causing a subject-tool contradiction.

Towards the contradiction, Yan chose one critical suggestion of the long feedback as later focus and crafted a prompt 'Give me a real-life example where taking risks has more advantages than disadvantages,' explaining this prompting as follows:

'At the first stage, I just had one movie example, but nothing about real life. I did know that the argument was not strong enough. That's why when ChatGPT asked me

to provide some real-life examples, I thought, I should do that. And that's why I asked it to give me some real-life examples'.

According to the quote, she already had a prior evaluation that her current example was insufficient, prompting her to recognise one of ChatGPT's suggestions on enriching evidence. Based on her previous argumentative knowledge, Yan incorporated her argument that 'taking risks has more advantages than disadvantages' into the prompt. Her action showed her appropriation of prompts and prior knowledge as mediating resources in solving the contradiction. The second strategy was generalised to indicate the two mediating resources, extending Principle 2.

Appropriation strategy 2-2: defining format and criteria in prompts. Towards a similar subject-tool contradiction caused by ChatGPT's vague feedback, another strategy was generalised to elicit concrete feedback by specifying the format and criteria in prompts. After receiving ChatGPT's evaluation for proofreading, Li and Wang did not accept the feedback because 'The advice it gives is very vague. It will say that your grammatical accuracy can be improved, but it won't say where you have problems'. (Li) and the format was not reader-friendly, as said by Wang: 'The format was a bit of a headache to look at. It puts the original sentence up, and then it has a correction'.

As visualised in Figure 7, writers needed actionable feedback, while ChatGPT's feedback was vague and in an inaccessible format, leading to a subject-tool contradiction. However, the two participants did not take further action at this stage, which could be understood as unsettled contradictions. Later, Li offered some clues for solving the contradictions through revising prompts:

'You can add the word count in that prompt. If there is a word requirement, it won't be too long... Sometimes it will change your correct one to another correct one, just to paraphrase. Maybe you can tell it in your prompt not to paraphrase. You just tell it, 'You don't need to revise until I make a grammatical mistake,' which is much more straightforward'.

In the excerpt, Li offered two possible solutions: specify the word count in prompts and ask ChatGPT to avoid unnecessary paraphrasing, both of which resonated with the principle of prompting ChatGPT with specificity.

Driven by the unsettled contradictions and participants' thoughts, some potential solutions were derived from literature to intervene in writing activities: asking ChatGPT to present revisions in a table (Mizumoto & Eguchi, 2023) and adding

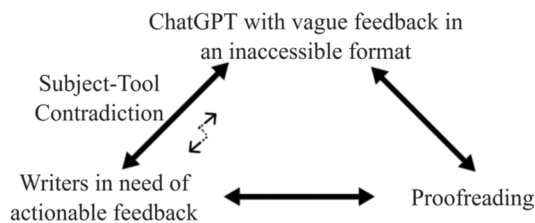


Figure 7. Subject-tool contradiction at proofreading stage (when writers need actionable feedback but meet ChatGPT's vague and inaccessible feedback).

more linguistic features such as the detailed IELTS band descriptors and categorisations of grammar errors from Han and Hyland (2015) and Ferris (2006). The third strategy on specifying the format and criteria in prompts was then generalised to more strategically employ prompts as mediating resources.

Design principle 3: integrate ChatGPT's response with criticality

The third principle, a newly-emerged one in Iteration 1, involves critically evaluating and incorporating ChatGPT's feedback into writing. Although the second principle focuses on prompting ChatGPT to give concrete responses, the aforementioned subject-tool contradiction could persist when ChatGPT generates another overly long text in response. Solving this contradiction led to the emergence of the third principle.

Appropriation strategy 3-1: trimming and extending based on prior knowledge. This strategy was to address the overly long feedback from ChatGPT by trimming unnecessary points and extending relevant ones. For instance, the exhaustive feedback posed barriers to Yan, as said by her: 'ChatGPT gave me a whole story of Steve Jobs and Apple and listed all the advantages and disadvantages, which I think was not that necessary for my essay'. Thus, ChatGPT gave Yan an example of Steve Jobs in every detail, though she expected concise and relevant evidence from ChatGPT.

Displayed in Figure 8 as a subject-tool contradiction, ChatGPT's response in this case mismatched with writers' needs for an appropriate example ready to use. Subsequently, Yan critically incorporated the ChatGPT-generated example, explaining how she trimmed the long feedback and extended it to serve her needs:

'I only need to give one or two (disadvantages), so I decided that the idea of financial loss and reputation is more important or more well-known...After providing an example, it's important to add one or two sentences to relate the example back to the argument or topic that you're writing'.

Accordingly, drawing on her argumentative writing knowledge, she retained two points to support her argument and added two sentences to link the ChatGPT-provided evidence with her argument. The resolution of the contradiction should be attributed to her appropriation of prior knowledge as the mediating resource to trim and extend ChatGPT's feedback, which was generalised as the fourth strategy.

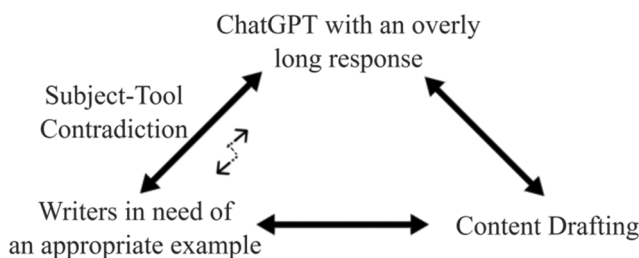


Figure 8. Subject-tool contradiction at content drafting stage (when writers need appropriate examples but meet ChatGPT's overly long response).

Appropriation strategy 3–2: comparing and noting down useful information. Facing a similar subject-tool contradiction due to ChatGPT's overly long response, Wang compared ChatGPT's content and hers and noted down useful ideas that resonated with her prior knowledge, explaining that:

'I compared these items. One was to compare my own wording, and the other was that if it was a new idea, I could selectively integrate it into my outline. If I felt that a particular point did not match well with my personal experience or understanding, I would not use it. Instead, I would select aspects that aligned closely with my own beliefs and applied them to my essay'.

Thus, Wang mixed ChatGPT's response with hers by comparing and noting down useful information. Her appropriation of prior knowledge and notes could promote her noticing of the current gap between her own writing and the ChatGPT-generated one, leading to the solution of the subject-tool contradiction and the emergence of the fifth strategy.

Design principle 4: adapt ChatGPT to different writing stages

The fourth principle is to appropriate ChatGPT as a brainstorming partner to generate sufficient ideas for outline generation and as a writing evaluator to offer focused suggestions for proofreading. The two strategies arose from a subject-object contradiction when writers lacked some inspiration before writing and a subject-tool contradiction when ChatGPT offered overly general feedback after writing.

Appropriation strategy 4–1: brainstorming with ChatGPT before writing. During outline generation, Wang sometimes met difficulties in generating adequate ideas, stating that: 'Because I may not know much about the topic, coming up with ideas is the hardest part for me'. As visualised in Figure 9, Wang was unfamiliar with the topic and could not develop sufficient content for the writing task, during which a tension between writers and the envisioned object arose.

Facing this contradiction, Wang introduced ChatGPT into the activity system to access abundant information as additional mediating resources. She described her strategy as below:

'The two most important questions I asked it were when I presented the original question to it...I intended to directly ask it from its knowledge base, so it could give me many points directly'.

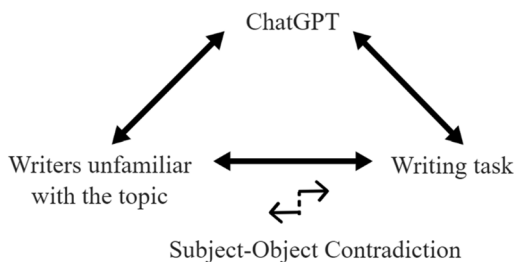


Figure 9. Subject-object contradiction at outline generation stage between writers unfamiliar with the topic and the writing task.

Wang directly sent the writing question to ChatGPT, leveraging ChatGPT's extensive knowledge base to gather diverse points and solve the contradiction. Appropriating ChatGPT for brainstorming could thus afford the resolution of subject-object contradiction, which was generalised as a strategy under the fourth principle.

Appropriation strategy 4–2: gaining focused feedback from ChatGPT after writing. This strategy was generalised in resolving an aforementioned contradiction: as shown in Figure 7, at the proofreading stage, a subject-tool contradiction arose because ChatGPT's vague response failed to meet writers' need for actionable feedback. Besides the aforementioned strategy to define criteria and format in prompts, participants also chose specific writing dimensions to ask for focused feedback. For example, ChatGPT gave Yan a comprehensive evaluation on the four IELTS writing dimensions. Regarding the grammar, ChatGPT indicated that there were some awkward phrases with one example sentence, but did not specify how to revise. Yan crafted two prompts respectively on how to rephrase this example sentence and rewrite other awkward phrases, followed by ChatGPT's specific revisions. She stressed the utility of ChatGPT's feedback:

'It's quite useful because as a second language writer, sometimes we are not very aware of what sentence is constructed awkwardly...Therefore, when we have the tool ChatGPT, it works like a critical friend, the person who is quite more fluent in English than me'.

Thus, Yan referred to ChatGPT as a more advanced peer critically evaluating problems and providing language scaffolding for L2 writers. Her actions of asking the two focused questions not only confirmed the second strategy of asking for elaboration, but the use of ChatGPT as an evaluator to gain focused feedback, through which strategy 4–2 was generalised.

To summarise, in Iteration 1, two initially-formed design principles were enriched, and two new principles were constructed from the mediated appropriation process. Participants have employed six appropriation strategies to solve within-tool, subject-object, subject-tool, and tool-object contradictions, driving forward the activity system and leading to revisions of the intervention.

Iteration 2: refinement

In Iteration 2, the second-version intervention was widely regarded as largely improved than the first version. However, some previously formed strategies did not suffice in solving contradictions. This section focuses on two new appropriation strategies that were adopted by participants and categorised into previous design principles. The results from the second iteration led to some refinement of the intervention, activity system, and design principles.

Appropriation strategy 1–2: combining search engines for information verification

This strategy was utilised when a tool alone might not suffice in providing reliable information. For example, facing a similar sign-object contradiction between writers' limited lexical resources and the writing task as aforementioned in Strategy 1–1,

writers employed search engines to access diverse platforms to compare and gain trustworthy information. Specifically, to find a technological term ‘挤兑’ (run-on), Zhang first used Google Translation but was skeptical about the answer. He then accessed Bilibili (a Chinese social media video platform) through Google to verify the translation of the term in a video. He finally decided to adopt the wording because it had been verified from two different sources accessed *via* search engines, as stated by him: ‘The reason I used it (this expression) was because the machine translation and the blogger (in the video) said so. All of them said so, so I thought it was more credible’. His action in solving the sign-object contradiction was later summarised as a strategy of using search engines, extending the first design principle.

Appropriation strategy 1–3: using AWE tools for grammar check

This strategy was generalised because ChatGPT showed obvious limitations in grammar check while AWE tools were more accurate. An exploratory revision from Iteration 1 was to incorporate the categories of grammatical errors into the prompt to elicit ChatGPT’s grammar evaluation. However, this was problematic in Iteration 2 through the manifestation of a tool-object and a within-tool contradiction. Firstly, ChatGPT once claimed that Li misused the word ‘cover’ which he did not use in writing, as explained by him, ‘I just kept searching for ‘cover.’ I couldn’t find ‘cover’... It seems that this is made up’. ChatGPT thus fabricated a grammar error on his use of ‘cover’.

As visualised in Figure 10, ChatGPT fabricating the grammar error could not serve the writers’ goal of grammatical accuracy during the proofreading stage. This fabrication could relate to the artificial hallucination, a situation of a machine generating seemingly ‘realistic information that do not correspond to any real-world input’ (Alkaissi & McFarlane, 2023, p.3).

Moreover, visualised as a within-tool contradiction in Figure 11, ChatGPT and Grammarly once provided contradictory grammar evaluations on whether to use a definite article.

Explaining that ‘ChatGPT told me to delete it (the article). Grammarly told me to add it. I chose to trust Grammarly more because my first instinct was to add it’, Zhang made the decision possibly based on his knowledge of grammar rules, leading to the resolution of the contradiction. Furthermore, a quick review of literature confirmed the advantage of AWE tools over ChatGPT regarding grammar evaluation: because of its generative nature, ChatGPT shows a tendency of

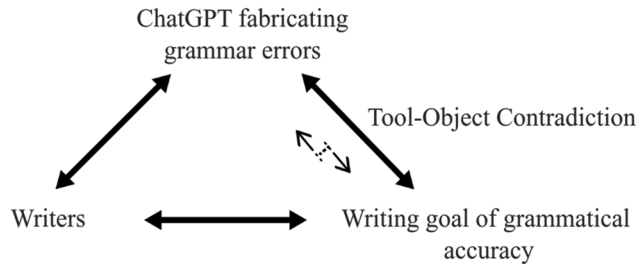


Figure 10. Tool-object contradiction at proofreading stage (when ChatGPT fabricates grammar errors and cannot meet the writing goal of grammatical accuracy).

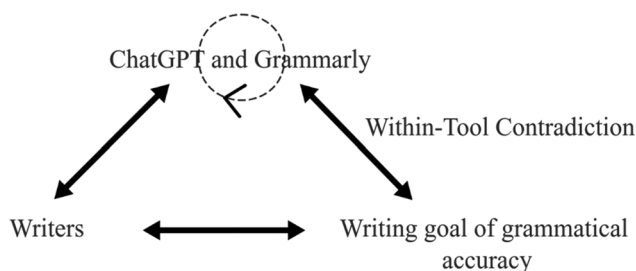


Figure 11. Within-tool contradiction at proofreading stage between ChatGPT and grammarly.

over-correction, while AWE tools show more accuracy than ChatGPT in grammar check (Wu et al., 2023; Xiao & Zhi, 2023). Therefore, a strategy of using AWE tools for grammar check was generalised under the first principle on orchestrating different tools.

Iteration 3: product

After revisions in the first two iterations, participants in Iteration 3 expressed that there was little need to revise the intervention, indicating that it was appropriate to conclude the research. Driven by contradictions and participants' solution efforts, the final six-element activity system has thus been developed (Figure 4), underpinning a final four-principle ChatGPT-assisted EFL writing framework (Figure 12). The design principles were developed both inductively, in generalising participants' strategies of solving contradictions and deductively in implicitly attending to different mediating resources. These principles, comprising two appropriation strategies each, are not mutually exclusive, but interrelate with each other to guide ChatGPT-assisted EFL writing.

Discussions

The research findings revolve around the iterative development of design principles for ChatGPT-assisted EFL writing, underpinned by the activity system framework. Through three research cycles, the study took a micro lens to reveal how different mediating factors come into play and result in contradictions. Through appropriating mediating resources to solve the four contradictions, participants exhibited their agency in reshaping their activity systems and constructing a culturally new pattern of the activity (Engeström, 2001). A four-principle framework has been developed to enact this new and optimal mode of ChatGPT-assisted activity.

The findings on contradictions generally concur with the potential applications of ChatGPT in EFL writing as discussed in existing research. The findings demonstrate how writers appropriate ChatGPT to resolve the sign-object contradiction arising from writers' limited lexical resources and the subject-object contradiction due to writers' unfamiliarity with the topic. This has shown ChatGPT's affordance in overcoming barriers in language proficiency and content generation, corresponding to previous evidence on the tool's scaffolding for language resources and idea development (Guo et al., 2023; Su et al., 2023; Zou & Huang, 2023). Meanwhile, some

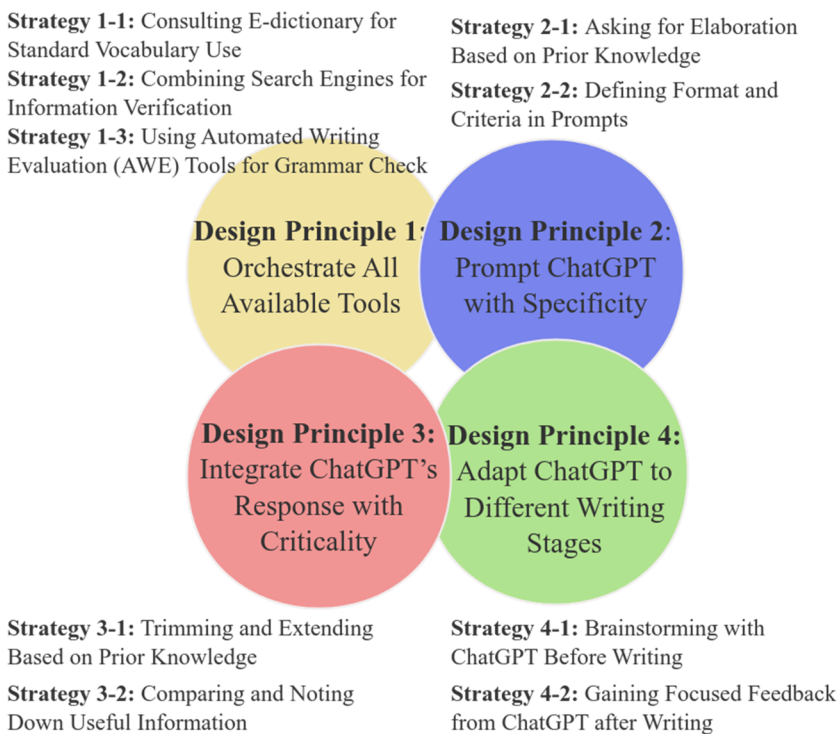


Figure 12. A four-principle framework for ChatGPT-assisted EFL writing.

contradictions have emerged between ChatGPT and writers' needs, the objectives of tasks, or other similar tools. These contradictions can be traced to ChatGPT's limitations, such as overly long and vague responses and fabrication of information, as widely discussed in previous research because of the tool's generative nature (Chan & Hu, 2023; Su et al., 2023; Warschauer et al., 2023).

However, findings in this study differ from previous ones on the benefits and limitations of ChatGPT in at least two ways. Firstly, the research posits the affordances and constraints of ChatGPT not in its abstract properties, but in its interplay with other mediating factors in the activity system. The findings have shown ChatGPT's affordances in a dynamic process where writers appropriate ChatGPT to solve subject-object and sign-object contradictions regarding the language and content aspects of writing. The constraints of ChatGPT have been understood in how it contradicts the subject, object, and similar tools in the activity system. Therefore, through the lens of contradictions, the tool's affordances and constraints could be reconceptualised as potential interactions among the mediating factors and are context-dependent (Lier 2010). Secondly, a micro lens on contradictions and solution efforts in this research could spotlight human agency in AI use: encountering the contradictions, writers did not only use ChatGPT reactively, but proactively adapted its outputs to suit their specific needs in local contexts. Writers' exert various appropriation strategies to maximise the potential of ChatGPT in EFL writing and navigate the tool's constraints, which echoes a call for human-centered AI use (Warschauer et al., 2023; Watson, 2023; Wu et al., 2023).

As for the four principles, they also share some commonalities with suggestions from previous studies. Specifically, the first principle ‘Orchestrate all available tools’ recommends writers mobilise the intellects of all available tools instead of relying on ChatGPT. Within ChatGPT-assisted writing research, researchers have compared ChatGPT against website-based resources (Zhang et al., 2023) or teacher feedback (Guo et al., 2023), framing ChatGPT as an alternative to current tools. Differently, this principle suggests writers employ ChatGPT as a supplementary tool and coordinate and combine different digital tools. This also resonates with suggestions from previous writing strategies research to use various tools such as the Internet, dictionaries, and textbooks (Kessler, 2020; Lei, 2008).

The second principle ‘Prompt ChatGPT with specificity’ centers on the unique aspect of ChatGPT - prompting - which can largely influence the quality of ChatGPT’s output (Warschauer et al., 2023; Watson, 2023). Crafting appropriate prompts is a trial-and-error process where users should iteratively refine prompts to make ChatGPT produce desired outputs (Dang et al., 2022). Through three rounds of research, we have included specific prompts regarding the writing criteria, corresponding with many researchers’ suggestions on providing evaluation rubrics to tailor ChatGPT’s feedback (Su et al., 2023; Warschauer et al., 2023; Watson, 2023). The iterative refinement of prompts also responds to Cress and Kimmerle (2023) call for researching an optimal human-AI interactive prompting process.

The third principle, ‘Integrate ChatGPT’s response with criticality’ spotlights human criticality and agency, further placing ChatGPT at a subordinate position in EFL writing (Zou & Huang). Warschauer et al. (2023); Watson (2023) have similarly recommended strategies including employing one’s previous experience and knowledge and considering the contextual factors such as writing purpose and genre requirements. These strategies under the third principle relate to the rule-mediated, tool-mediated, and object-mediated aspects of writing activities, emphasising the importance of human intellectual and contextual awareness.

The fourth principle, ‘Adapt ChatGPT to different writing stages’, based on the evolving division of labor at different writing stages, shows ‘when to introduce what functions of AI tools’ (Warschauer et al., 2023; Watson, 2023, p. 5). The first and second appropriation strategies under this principle correspond with previous suggestions on using ChatGPT for brainstorming (Berg, 2023; Chan & Zhou, 2023) and gaining writing evaluation (Barrett & Pack, 2023; Mizumoto & Eguchi, 2023). Merely using ChatGPT as a language resource during content drafting may avoid the textual borrowing and plagiarism that are currently controversial both in research and practice (Su et al., 2023; Warschauer et al., 2023).

More importantly, different from previous studies on the applications of ChatGPT and suggestions, the four-principle framework is theoretically underpinned by a culturally new activity system and practically designed with actionable guidelines. First, we position ChatGPT in its interplay with other mediating factors and show how their contradictions could drive expansive developmental transformations. Focusing on ChatGPT’s utility, previous studies discussed ChatGPT in its physical property and posited the learning process as stable and linear where subjects acquire some skills, which, according to Engeström (2001), could be based on standard theories of learning. By contrast, Engeström argues for expansive learning, a new

approach where subjects could learn a new form of activity as they are creating the activity. In this research, the initial-version activity system proceeded through three iterative cycles of expansive learning and was transformed into a culturally new pattern of ChatGPT-assisted writing activity. Human agency manifested itself in the process, as participants were able to reshape their activity systems through solving contradictions (Engeström, 2001). Thus, from the reconstruction of the activity system and resolution of contradictions, four design principles and nine appropriation strategies have emerged, guiding the enactment of the iteratively developed and culturally new activity. The first three principles can strategically employ ChatGPT, other available artifacts, and rules as mediating resources, and the fourth principle attends to the contextual division of labor in the ChatGPT community. Thus, underpinned by an activity system, the four-principle framework relates to the rule-mediated, tool-mediated, object-mediated, and division-of-labor-mediated aspects of writing activities, spotlighting human agency.

Secondly, the four-principle framework is developed with an iteratively refined intervention with detailed use suggestions, while most of the previous studies on ChatGPT-assisted writing did not go into details with empirical evidence (Su et al., 2023; Warschauer et al., 2023; Watson, 2023; Zou & Huang, 2023). For example, in Zou and Huang's (2023) research, EFL students suggested five general ways for ChatGPT-assisted writing, but were uncertain about how to write prompts and use ChatGPT as a self-learning tool. Students thus expected relevant and specific guidance. In the present study, the design principles have been iteratively designed with actionable guidelines based on writing practices. Moreover, similar to the writing strategies identified by Lei (2008) that were found to be interrelated and goal-oriented, the strategies in this study were not randomly used by writers, but aimed to address certain contradictions. Resolving these contradictions required not just a single strategy, but a coordinated interaction among strategies within and across the four categories. Writers in this research used various strategies interconnectedly, as observed in other L2 learning studies as 'orchestration' (Anderson, 2005) and 'strategy chain' (Oxford, 2001).

Conclusion

This research has iteratively developed an innovative intervention, an AT framework, and a four-principle framework with nine strategies for ChatGPT-assisted EFL writing. The developmental trajectory of design principles is underpinned by the AT framework, which together explain how and why the final intervention could function in practice. The findings can shed some light on the emerging field of AI-assisted writing in four ways. First, this is one of the first studies to conceptualise the ChatGPT-assisted writing process and strategies through the lens of AT. AT has been further advanced to show the transformative influence of ChatGPT in causing contradictions, writers' agency in appropriating ChatGPT to solve contradictions, and the expansive transformation of the activity system. Second, the four-principle ChatGPT-assisted EFL writing framework can contribute to future domain-specific AI literacy frameworks. As predicted by Davenport and Mittal (2022), we may only scratch the surface functions of GenAI, which spotlights the need to achieve more in-depth and context-relevant

mastery of GenAI. While some multidisciplinary and holistic AI literacy frameworks (e.g. Allen & Kendeou, 2024) have emerged recently, our proposed framework can guide students' strategic use of AI especially in EFL argumentative writing. Third, this is also the first attempt in the field to bring together AT and DBR, showing the two as a natural pair since the developmental process of activity systems resonates with the iterative cycle of DBR. Fourth, through DBR, we have iteratively optimised prompts for Large Language Model (LLM)-based chatbots and developed context-sensitive interventions, which could inform self-regulated learning, EFL writing instructions, and the fine-tuning of AI writing assistance technologies. For instance, regarding the application of the first design principle, teachers can encourage writers to mobilise other available tools rather than merely ChatGPT in EFL writing. Educational product designers can consider embedding e-dictionaries or other tools into future AI conversational systems. For self-regulated learners, the third design principle can serve as a reminder for learners to critically evaluate AI-generated responses. To support the practical implementation of design principles 1, 2, and 4, we provide specific instructions in the [Appendix C](#), which were part of our intervention.

However, the study is not without limitations. This study was exploratory in nature, and the sample size was small. The participants' high proficiency levels and familiarity with digital tools could introduce possible biases, and the findings may not be totally applicable to less proficient learners with limited digital literacy or different writing tasks. Practitioners should carefully examine contexts when transferring this case study's findings (Dooley, 2002). Special attention should be paid to adapting this framework for learners with lower AI literacy. For instance, Design Principle 3 can be supported by explicit classroom discussions or evaluation rubrics to help learners critically assess AI-generated output. Future research could recruit a larger number of EFL students with various levels of English proficiency to make a statistical generalisation of findings. Meanwhile, our analysis relied largely on self-reported data, which may be subject to biases such as social desirability. Though we triangulated multiple data sources including screen recordings and chat logs, future studies could benefit from incorporating additional real-time measures (e.g. electroencephalograms, eye-tracking) to capture cognitive activities as suggested in Järvelä et al. (2023). Thirdly, this study focused on contradictions relevant to ChatGPT, while there could also be contradictions within and across other elements of writing activities (Li, 2013). A more comprehensive picture of the writing activities through the construct of contradictions could be explored. Finally, this research concentrated on ChatGPT 3.5, while future research could explore more advanced GenAI tools to further 'tame the beast of AI'.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix A. Interview questions

Semi-Structured Interview on Intervention Revision

How do you feel about the writing guideline in general?

Do you have any suggestions for improving the part on outline generation? Content Drafting? And proofreading?

Semi-Structured Interview on Activity Framework Revision

Q1: What do you think are the rules to write the essay in a ChatGPT-assisted way?

Q2: What were your goals while writing the essay?

Q3: How did you use ChatGPT to complete the writing task?

Q4: What were some tools or resources you used when interacting with ChatGPT and performing the writing task? Why did you use them, and how did them help/constrain you in completing the essay?

Q5: What do you think is the role of ChatGPT in the writing process? How do you understand your relationship with ChatGPT during writing?

Q6: Do you have any other comments or thoughts to share?

Stimulated-recall interview on contradictions and solutions

Then, can you point out if there is any moment that you feel you encountered some contradictions? I will also share my screen with you to discuss some specific moments that you might encounter some contradictions.

Can you explain what kind of contradictions, challenges, or difficulties you encountered in this screenshot?

Can you recall why did you decide to ask this follow-up question/take this action?

Can you recall how did you evaluate this response from ChatGPT?

How did you incorporate its response into your writing?

Why did you decide to use e-dictionary/Grammarly/search engines at this moment?

Appendix B. Evolution of the intervention

The following tables present the revisions made to our first-version and second-version interventions across the first two research rounds, with no revisions in the third round.

Iteration 1:

Writing Stages	Revisions to the First-Version Intervention with Screenshots of the Second-Version Intervention	Corresponding Appropriation Strategies
Outline Generation	A step to 'brainstorm with ChatGPT' was added with suggested prompts. 1.(Optional) Brainstorm with ChatGPT by asking about the explanation of key concepts or the writing topic. "Provide a short definition of XXX (the keywords in the writing task) " "What is the nature of XXX? Give me a short answer." Or: You could send the writing question to ChatGPT after you have some initial ideas A step to 'compare and evaluate ChatGPT-generated outline with yours' was added with a prompt asking ChatGPT to independently generate an outline instead of providing feedback on writers' outlines. 3.Compare and evaluate ChatGPT-generated outline with yours "Act as an excellent argumentation outliner who follows the following structure and criteria to answer an IELTS Task 2: IELTS task 2: (Please insert the question here) Structure: Main claim: (Please insert your main claim here) Sub claim 1: Evidence 1: Reasoning 1: Sub claim 2: Evidence 2: Reasoning 2: Criteria: the sub-claims are closely connected and progressed logically to support the main claim the evidence can support the sub-claims the reasoning provides a justification of why the evidence support the sub-claim"	Strategy 4-1: Brainstorming with ChatGPT before writing The previous prompt to gain feedback on writers' outlines was problematic since ChatGPT turned out to generate vague response (see more detailed discussion on this in Appropriation Strategy 3–2). Strategy 3–2: Comparing and Noting Down Useful Information
Content Drafting	A suggestion on using e-dictionaries for authentic word-searching was added. ➤ You are encouraged to use e-dictionary to seek vocabulary and its authentic use, and any other available resources are encouraged to be used.	Strategy 1–1: Consulting E-dictionary for Standard Vocabulary Use
Proofreading	The proofreading stage was divided into two phases: general evaluation and focused feedback. Stage 3-Reflection and Proofreading During this stage, you are advised to first gain comprehensive and general feedback on the four writing criteria, with specific scoring from ChatGPT. Please note that the scoring from ChatGPT cannot predict your actual scores in exams, but serves as a reflective tool for you to improve your writing. After gaining an overall evaluation, you can then choose to focus on some specific dimensions, ask for focused feedback, and revise your writing. 1. Phase 1: General Evaluation 2. Phase 2: Specific Evaluation and Proofreading You can ask ChatGPT to specifically provide feedback on any of the four dimensions and then revise your writing.	Strategy 4–2: Gaining focused feedback from ChatGPT after writing

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Writing Stages	Revisions to the First-Version Intervention with Screenshots of the Second-Version Intervention	Corresponding Appropriation Strategies
	Detailed writing band descriptors and expected format were added into the prompt for general evaluation. The element of scoring respectively on four IELTS writing dimensions was added to elicit more specific feedback, with a reminder of ChatGPT's limited reliability. 1. Phase 1: General Evaluation "Act as a professional IELTS writing tutor who provides critical and specific feedback to the writing on the four dimensions of IELTS writing criteria according to the band descriptors. Please give me individual scores of each of the four dimensions, how does my writing match or mismatch the band descriptors, and ways to improve each dimension. Band Descriptors: (1) Task Response Band 8: . sufficiently addresses all parts of the task . presents a well-developed response to the question with relevant, extended and supported ideas Band 7: . addresses all parts of the task . presents a clear position throughout the response . presents, extends and supports main ideas, but there may be a tendency to <i>over-generalise</i> and/or supporting ideas may lack focus Band 6: . addresses all parts of the task although some parts may be more fully covered than others . presents a relevant position although the conclusions may become unclear or repetitive . presents relevant main ideas but some may be inadequately developed/unclear The categories of grammatical errors were specified in prompts. Some prompts that have been used by participants for focused feedback on grammar and lexis were included. 2. Phase 2: Specific Evaluation and Proofreading You can ask ChatGPT to specifically provide feedback on any of the four dimensions and then revise your writing. For Task Response and Cohesion and Coherence, you are advised to ask follow-up questions for clarification, explanation, and correction directly according to <u>ChatGPT</u>'s evaluation. Here are some suggested ways: "<i>Point out how...</i>" "<i>Give me concrete examples of ...</i>" For Grammar Range and Accuracy and Lexical Resource, you are also encouraged to improve the sentence structure and word choice. Here are some suggested ways: "<i>Did you notice any repetitive use of specific words in this writing? If so, point out and show me how to improve.</i>" "<i>Can you suggest more varied and advanced word choices to allow more lexical flexibility and diversity?</i>" "<i>Can you suggest more varied and complex sentence structures to allow some flexibility and diversity? Please do not paraphrase my original sentences. Avoid overly complex and formal words.</i>"	Strategy 2–2: Defining format and criteria in prompts Strategy 2–2: Defining format and criteria in prompts

Iteration 2

Writing Stages	Revisions to the second-version intervention with screenshots of the third-version intervention (See the bold parts in the screenshots)	Corresponding Appropriation Strategies
Outline Generation	The criteria for ChatGPT-generated examples were made clearer as: 'The evidence can involve real-world examples to support the sub-claims' 5. Phase 5: Compare and evaluate ChatGPT-generated outline with yours "Act as an excellent argumentation outliner who follows the following structure to answer an IELTS Task 2: IELTS task 2: (Please insert the question here) Structure: Main claim: (Please insert your main claim here) Sub claim 1: Evidence 1: Reasoning 1: Sub claim 2: Evidence 2: Reasoning 2: Criteria: The sub-claims are closely connected and progressed logically to support claim; The evidence can involve real-world examples or expert quotes to support sub-claims the reasoning provides a justification of why the evidence support the sub-cl	Strategy 2–2: Defining format and criteria in prompts

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Writing Stages	Revisions to the second-version intervention with screenshots of the third-version intervention (See the bold parts in the screenshots)	Corresponding Appropriation Strategies
Content Drafting	A suggestion on prompting ChatGPT in L1 to gain vocabulary support at content drafting stage was added. Stage 2-Content Drafting ➤ During the 1-3 phases, you can ask questions to ChatGPT but cannot ask it to write paragraphs for you. ➤ You are encouraged to use e-dictionary to seek vocabulary and its authentic use, and any other available resources are encouraged to be used. ➤ You could also ask ChatGPT in your L1 to search for vocabulary. Search engines could be useful particularly in searching for technical terms. ➤ You should not totally copy or paste contents from ChatGPT or other resources. ➤ The following structure comes from a famous and experienced IELTS teacher and previous examiner Simon (Please note that this is for reference, and you can structure your writing specifically regarding the question)	Strategy 1–2: Combining search engines for Information Verification
Proofreading	More suggested prompts for focused feedback were added. 2. Phase 2: Specific Evaluation and Proofreading You can ask ChatGPT to specifically provide feedback on any of the four d and then revise your writing. For Task Response and Cohesion and Coherence, you are advised to ask questions for clarification, explanation, and correction directly acc ChatGPT's evaluation. Here are some suggested ways: "Point out how..." concrete examples of..." "Where can I add cohesive devices in my writing can I strengthen my reasoning in XXX?"	Strategy 4–2: Gaining Focused feedback from ChatGPT during Proofreading

Appendix C. Instructions for ChatGPT-assisted EFL writing with reference to design principles

Design principle 1: Orchestrate all available tools

Strategy 1–1 Consulting E-dictionary for standard vocabulary use

- You are encouraged to use e-dictionary to seek vocabulary and its authentic use, and any other available resources are encouraged to be used.

Strategy 1–2 Combining search engines for information verification

- You are encouraged to use search engines (e.g. Google, Baidu) to check and enrich the evidence part of your outline

Design principle 2: Prompt ChatGPT with specificity

Strategy 2–1 Asking for elaboration based on prior knowledge

- You are encouraged to ask follow-up questions to ChatGPT for clarification

Design Principle 4: Adapt ChatGPT to Different Writing Stages

Strategy 4–1: Brainstorming with ChatGPT Before Writing

1. Brainstorm with ChatGPT by asking about the explanation of key concepts or the writing topic.

'Provide a short definition of XXX (the keywords in the writing task)'

'What is the nature of XXX? Give me a short answer.'

Or: You could send the writing question to ChatGPT after you have some initial ideas: 'What are the reasons or results/advantages or disadvantages/problems or solutions of XXX'

Strategy 4–2: Gaining focused feedback from ChatGPT after writing

1. Phase 1: general evaluation

'Act as a professional IELTS writing tutor who provides critical and specific feedback to the writing on the four dimensions of IELTS writing criteria according to the band descriptors. Please use bullet points to list the four things within each of the four dimensions: (1)

which band descriptors (6 or 7 or 8) does my writing best match, (2) how does my writing match with the descriptors of this band score, (3) how does my writing mismatch with the descriptors of the higher band score, and (4) ways to improve each dimension. Please refer to the specific and concrete content of my writing.

Band Descriptors:

(1) Task Response

Band 8:

- sufficiently addresses all parts of the task
- presents a well-developed response to the question with relevant, extended and supported ideas

Band 7:

- addresses all parts of the task
- presents a clear position throughout the response
- presents, extends and supports main ideas, but
- there may be a tendency to over-generalise and/or supporting ideas may lack focus

Band 6:

- addresses all parts of the task although some parts may be more fully covered than others
- presents a relevant position although the
- conclusions may become unclear or repetitive
- presents relevant main ideas but some may be inadequately developed/unclear

(2) Coherence and Cohesion

Band 8:

- sequences information and ideas logically
- manages all aspects of cohesion well
- uses paragraphing sufficiently and appropriately

Band 7:

- logically organises information and ideas; there is clear progression throughout
- uses a range of cohesive devices appropriately although there may be some under-/over-use
- presents a clear central topic within each paragraph

Band 6:

- arranges information and ideas coherently and there is a clear overall progression
- uses cohesive devices effectively, but cohesion within and/or between sentences may be faulty or mechanical
- may not always use referencing clearly or appropriately
- uses paragraphing, but not always logically

(3) Lexical Resource

Band 8:

- uses a wide range of vocabulary fluently and flexibly to convey precise meanings
- skilfully uses uncommon lexical items but there may be occasional inaccuracies in word choice and collocation
- produces rare errors in spelling and/or word formation

Band 7:

- uses a sufficient range of vocabulary to allow some flexibility and precision
- uses less common lexical items with some awareness of style and collocation
- may produce occasional errors in word choice, spelling and/or word formation

Band 6:

- uses an adequate range of vocabulary for the task
- attempts to use less common vocabulary but with some inaccuracy
- makes some errors in spelling and/or word formation, but they do not impede communication

(4)Grammatical Range and Accuracy

Band 8:

- uses a wide range of structures
- the majority of sentences are error-free
- makes only very occasional errors or inappropriacies

Band 7:

- uses a variety of complex structures
- produces frequent error-free sentences
- has good control of grammar and punctuation but may make a few errors

Band 6:

- uses a mix of simple and complex sentence forms
- makes some errors in grammar and punctuation but they rarely reduce communication

Here is the writing task:

Here is my writing:'

2. Phase 2: Specific Evaluation and Proofreading

You can ask ChatGPT to specifically provide feedback on any of the four dimensions and then revise your writing.

For **Task Response and Cohesion and Coherence**, you are advised to ask follow-up questions for clarification, explanation, and correction directly according to ChatGPT's evaluation. Here are some suggested ways: 'Point out how...' 'Give me concrete examples of ...' '**Where can I add cohesive devices in my writing?**' '**How can I strengthen my reasoning in XXX?**'

For **Grammar Accuracy and Lexical Resource**, you are advised to use Grammarly for grammar correctness and clarity, because ChatGPT is about understanding and generating language in a way that mimics human conversation, while Grammarly is more focused on analysing and improving the quality of written text.

You can also use ChatGPT or other tools for double check.

For **Grammar Range and Lexical Resource at a more advanced level**, you are also encouraged to improve the sentence structure and word choice. Here are some suggested ways: 'Did you notice any repetitive use of specific words in this writing? If so, point out and show me how to improve.'

'Can you suggest more varied and advanced word choices to allow more lexical flexibility and diversity?'

'Can you suggest more varied and complex sentence structures to allow some flexibility and diversity? Please do not paraphrase my original sentences. Avoid overly complex and formal words.'